Inference Pipeline: Species and Individual Identification

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Overview

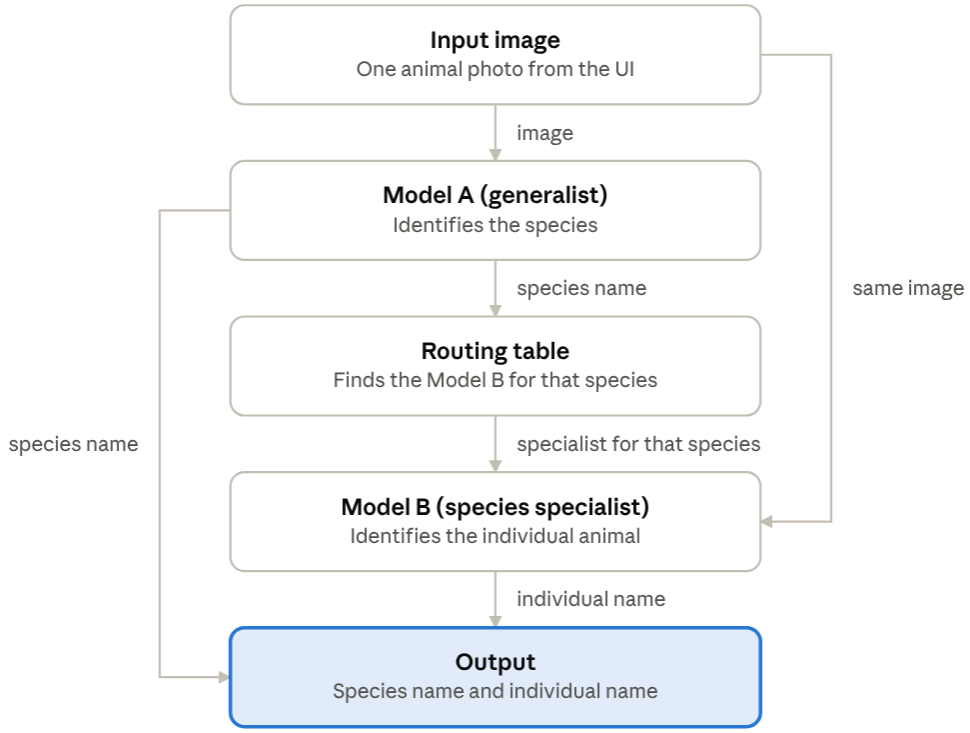
The system takes one animal image and returns two things: the species and the individual animal. Model A identifies the species, a routing table finds the Model B specialist for that species, and that specialist identifies the individual.

The same input image is used in both stages. This document describes the inference flow so the UI and back-end teams can build against it.

Pipeline steps

The input image is used twice: Model A reads it to find the species, then the matching Model B reads the same image to find the individual.

One image yields a species from Model A and an individual from Model B



inference flow · 5 steps

The species name from Model A goes to the routing table and to the output. The routing table returns the Model B for that species, and the original image goes straight to that model.

Components

Three parts make up the pipeline: one generalist model, one routing table, and one specialist model per species.

Component	Role	Input	Output
Model A (generalist)	Trained on all species together; identifies which species the image shows	Input image	Species name
Routing table	Database table that stores the Model B specialists and finds the one for a given species	Species name	The Model B specialist for that species
Model B (specialists)	One sub-model per species, each trained only on that species' individuals; identifies the individual animal	The same input image	Individual name

There are four species at present (dog, hyena, leopard, macaque), so four Model B specialists. Training data for Model A comes from the `Animal_All_Species` table, and each specialist trains on its own `species_<name>` table.

Output

The system returns the species name and the individual name for the submitted image. An example response is shown below; the field names and values are illustrative and still to be agreed between the teams.

```
{
  "species": "hyena",
  "individual": "individual_17"
}
```

Integration notes

The UI team sends one image in and shows two names back; the back-end team runs the two stages in between.

UI

- Let the user submit a single animal image.
- Display the species name and the individual name returned by the system.

Back end

1. Receive the image from the UI.
2. Run the image through Model A to get the species.
3. Look up that species in the routing table to find its Model B specialist.
4. Run the same image through that specialist to get the individual.
5. Return the species name and the individual name to the UI.

The image is used twice (Model A, then the specialist), so keep it available for the whole request.